higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CONCRETE STRUCTURES
NQF LEVEL 2

(12020132)

22 November 2019 (Y-Paper)
13:00–16:00

Calculators may be used.

This question paper consists of 5 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1: INTRODUCTION TO FORMWORK AND FALSEWORK

1.1 Define the following terms regarding working with concrete structures:

1.1.1 Formwork

1.1.2 Falsework

(2 × 3) (6)

1.2 List FOUR requirements of formwork for producing a smooth, flat and neat concrete finish. (4)

[10]

QUESTION 2: FORM FACE MATERIALS AND SPECIAL TYPES OF FORMS

By means of a diagram, show the following types of slumps:

- True slump (4)
- Shear slump (2)
- Collapse slump (4)

[10]

QUESTION 3: PRESSURE RESISTED BY FORMWORK AND FALSEWORK

3.1 Use a diagram to explain the lifting force of concrete in a conically shaped form. (7)

3.2 Make a neat labelled freehand sketch of a twisted wire loop wall tie. (4)

3.3 List FOUR factors that govern the pressure exerted by concrete on deep vertical forms. (4)

[15]

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QUESTION 4: ERECTING FORMWORK AND FALSEWORK

4.1 Make a neat vertical sectional drawing to illustrate the falsework and formwork for a concrete slab situated between two existing columns. The drawing must include the following detail:

- Sole plate
- Jacks
- Props
- Ledgers
- Bearers
- Decking
- Transom
- The two existing columns (12)

4.2 Define the following terms:

4.2.1 Ledgers

4.2.2 Bearers

4.2.3 Transoms

4.2.4 Girders

(4 × 2) (8)

4.3 Name FIVE uses of wedges. (5)
[25]

QUESTION 5: VERTICAL STOP ENDS, WATER AND JOINTS

Draw a neat freehand sketch to show the difference between the following joints:

5.1 Simple vertical construction joint

5.2 Joggle joint

5.3 Construction joint with starter bars

5.4 Vertical construction joint with starter bars

5.5 Rebate at a construction joint

(5 × 2) **[10]**

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QUESTION 6: USING RELEASING AGENTS

6.1 Releasing agents contain toxic solvents.

Describe FOUR precautionary measures to be taken when storing, handling and working with releasing agents. (4)

6.2 Name THREE instances where a releasing agent must be applied to a formwork face. (3)

6.3 Briefly describe the manufacturing process of chemical releasing agents. (3)

[10]

QUESTION 7: WORKMANSHIP AND DETAILING IN CONCRETE STRUCTURES

7.1 Make a neat labelled freehand sketch to show the effect of a rebate on a concrete cover. (8)

7.2 List 10 points that should be included on a formwork checklist. (10)

7.3 State TWO methods that could influence insufficient concrete cover. (2)

[20]

TOTAL: 100

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